Lightweight mannequins for TidyBot experiments

Checked October 11, 2026. Listed prices exclude US delivery unless noted.

Recommended starting points

- **Whole mannequin:** AFellow KM03-WEG, 94 cm. Articulated limbs; featureless head. Confirm mass and shoulder axes before ordering.
- **Lower-cost arm-focused setup:** Trade Lines 15776, 103 cm, **GBP £155 ex VAT**. Articulated wooden arms; fixed torso setup preferred.
- **Reproducible joint-limit experiments:** adapt the open-CAD Dummy Arm with adjustable stops. This requires fabrication.

Candidate shortlist

Product link	Height and mass	Listed price	Experimental fit
AFellow KM03-WEG	94 cm; mass unpublished	**Quote required**	First whole-body inquiry; arms/legs articulate; can sit.
Trade Lines 15776	103 cm; mass unpublished	**GBP £155 ex VAT / £186 inc VAT**	Arm-focused lead; wooden arms, hands and fingers. Leg articulation unconfirmed.
Store Supply 70370	~77 cm with head; mass unpublished	**USD $114.95**	Bendable limbs; removable head. Grasping/contact baseline.
VoiceBooster / TK child	96.5 cm; **3.40 kg including stand**	**USD $124**	Bendable frame; cannot sit. Sharp bends can fracture foam.
Kyoto Kagaku M73	70 cm; **~5.6 kg**	**Quote required**	Compact medical fallback; articulated metal frame under soft skin.
Kyoto Kagaku M86	98 cm; **~7.7 kg**	**EUR €5,003.47 inc VAT**	Medical fallback; heavier and expensive. Regional price.

Joint limits and appearance

No shortlisted commercial model has verified user-settable shoulder and elbow angular stops. Poseability or friction adjustment does not establish an adjustable physical endpoint.

Candidate	Joint limit and appearance notes
AFellow	Shoulder axes, elbow mechanism and stops unverified. Official photo: smooth white, featureless head.
Trade Lines	Shoulder axes and stops unverified. Matt white finish; facial detail unverified.
Store Supply / TK	Bendable frames are poor fixtures for learning repeatable rigid limits. Store Supply head is removable.
Kyoto M73 / M86	Articulated joints; adjustable stops unverified. M73 photo: realistic infant face. M86 face unverified.

No candidate is confirmed free of modeled genitalia. Obtain clear pelvis photos and written confirmation; clothed photos are insufficient.

Custom arm alternative

- **Dummy Arm paper** · **SolidWorks and STEP CAD:** three shoulder axes and hinge elbow; ~300/250 mm segments; reported masses 1.8/1.0 kg.
- **Proposed adaptation:** lighter links, padded contact surfaces and adjustable mechanical stops for shoulder flexion, shoulder abduction and elbow flexion. A face and pelvis can be omitted.
- **Cost:** paper reports **EUR €78.50 materials in 2024**, excluding labor. New stops, fixture and fabrication are additional; this is not a current complete-build quote.
- **Optional elbow prototype:** Brace Direct EBB239, **USD $79.99**, pre-order. Adjustable ROM stops; adult fit. Mannequin fit and robot-load rating unverified; shoulder stops still needed.

Related research

- **Manip4Care 2025:** real mannequin arm repositioning with UR5; physical shoulder had only **1 DoF**.
- **I-DRESS 2017: 1.85 m wooden mannequin** with spring-tensioned joints; held fixed poses during dressing trials.
- **DOROTHY:** adjustable, sensorized upper-limb research dummy; public purchase price unconfirmed.

Before ordering

- **Mechanism:** verify both shoulder motions, elbow hinge, accessible stops and independent friction adjustment.
- **TidyBot fit:** request body-only and individual arm-segment masses, dimensions, joint torque and repeated-loading limits. Start with the torso secured; check the actual robot arm/gripper capacity.
- **Appearance and quote:** confirm no modeled external genitalia, preferred head style, delivered US cost, lead time and return terms.